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SEAT No. :

PD-411

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**B.E. (MECHANICAL) (Insem.)
e-Vehicle Standards, Charging and Safety
(Honors in Electric Vehicles)
(2019 Pattern) (Semester - VIII) (302036MJ)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of Calculator is allowed.
- 5) Assume Suitable data if necessary.

Q1) a) Write a note on rechargeable electrical energy storage system. [7]

b) Explain electrical safety requirements of electric power train vehicle. [8]

OR

Q2) a) Write a note on Automotive Industry Standards (AIS). What are the types of AIS? [7]

b) Explain test procedure for measurement of electrical energy consumption. [8]

Q3) a) Explain testing standards for hybrid electric vehicle. [7]

b) Write a note on retrofitment standards on vehicles of L and N Category. [8]

OR

Q4) a) Write a note on method for measuring the electricity balance of the Battery of OVC and HEVS. [7]

b) Describe method of measuring the electric range of vehicles powered by a Hybrid electric power train. [8]

